

Qatar-1b

R-0429

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_180908 · created 2026-07-19T10:20:52+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458369.8186 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.057
Transit depth (Rp/Rs)^2	3.7 +/- 2.18 [%]
Orbital Inclination (inc)	89.07 +/- 2.88
Airmass coefficient 1 (a1)	0.44 +/- 0.24
Airmass coefficient 2 (a2)	0.48 +/- 0.48
Scatter in the residuals of the lightcurve fit is	57.97 %
Best Comparison Star	None
Optimal Aperture	2.41
Optimal Annulus	3.0
Transit Duration (day)	0.0833 +/- 0.0068

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.192 $\pm$ 0.057 vs 0.1463 (deviation 0.63 σ)
Duration fit vs published	0.0833 d vs 0.0692 d (deviation 1.63 σ)
Mid-time O-C	-1.9 min
Depth SNR	2.6
Residual scatter	57.97 %

Light curve

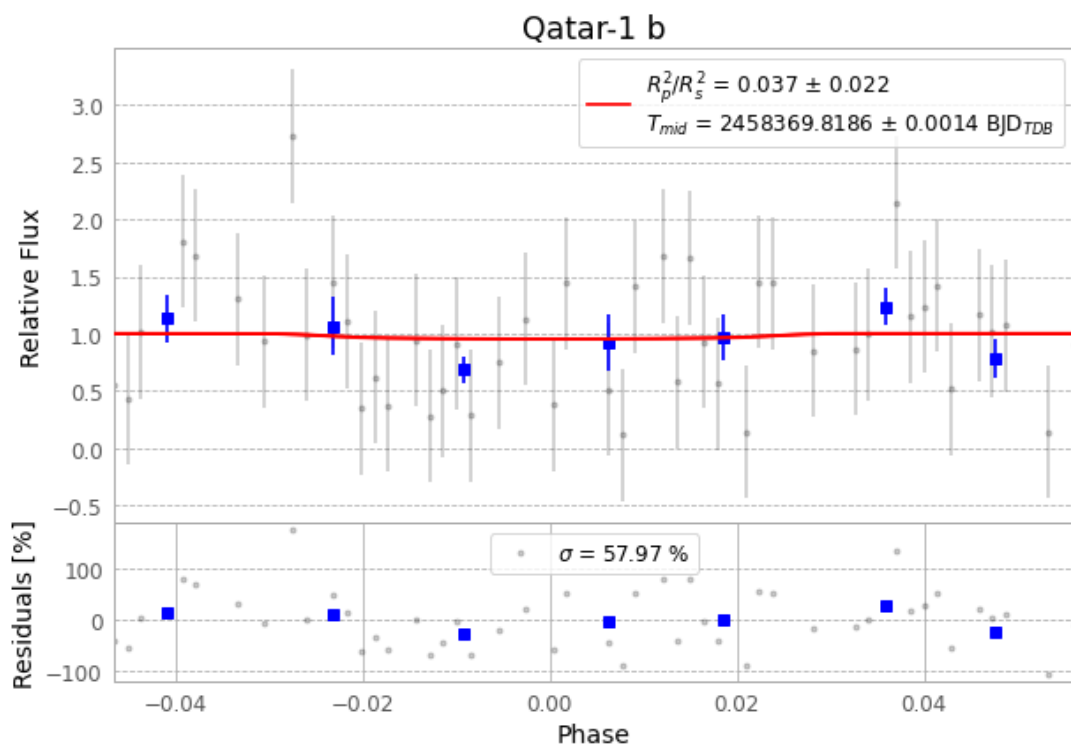


Figure 1 — FinalLightCurve_Qatar-1 b_2018-09-07.png (SHA-256 faa06a6b79202442...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [106, 407], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a0a2361a135d7fce...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

72 FITS frames (47 MB), Qatar-1180908055712.FITS → Qatar-1180908093012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-180908.log (8534786a0210...), FinalLightCurve_Qatar-1 b_2018-09-07.png (faa06a6b7920...), FinalLightCurve_Qatar-1 b_2018-09-07.csv (3faf3b3bc78e...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 10:20Z.